

CS 181 - Artificial Intelligence

Sibeï Yang
SIST, ShanghaiTech

Administrative Stuff

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Office hours: TBA



Administrative Stuff

▶ Classes

- ▶ Mon/Wed 8:15-9:55am @教学中心203
- ▶ 16 weeks
- ▶ Language: English/Chinese

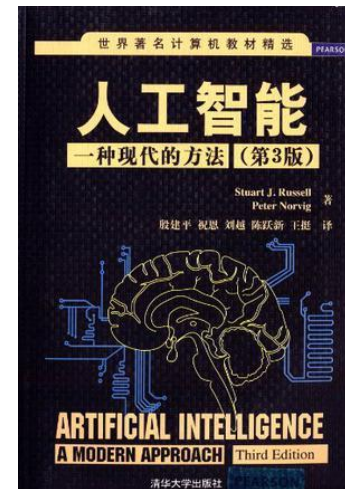
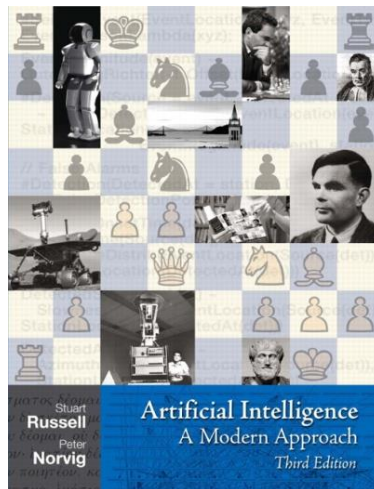
▶ Prerequisite

- ▶ Programming Languages, Data Structures, Algorithms
- ▶ Discrete Mathematics
- ▶ Probability and Statistics
- ▶ Linear Algebra
- ▶ Calculus



Administrative Stuff

- ▶ Main textbook
 - ▶ [AIMA] Artificial Intelligence: A Modern Approach, Stuart Russell and Peter Norvig, 3rd edition, 2009.
 - ▶ [中译版] 人工智能：一种现代的方法（第3版），2013



- ▶ Additional reference books may also be used
-

Administrative Stuff

- ▶ Blackboard
 - ▶ Announcements, homework assignments, slides, etc.
- ▶ Piazza
 - ▶ Discussion and QA
- ▶ AutoLab
 - ▶ Programming assignments
- ▶ Email the teaching team (**ALL**) in **a manner**
 - ▶ Subject:
[CS181] Assignment#x/Programming#x/Lecture#x/Project/...
- ▶ Roster will be updated after the 2nd week



Administrative Stuff

- ▶ Grading (tentative)
 - ▶ 6 homework assignments (10%)
 - ▶ 6 programming assignments (25%)
 - ▶ Project (15%): 2nd half of the semester
 - ▶ Midterm exam (25%): in mid April
 - ▶ Final exam (25%): in mid June



Administrative Stuff

▶ Plagiarism

- ▶ All assignments must be done individually
 - ▶ You may not look at solutions from any other source
 - ▶ You may not share solutions with any other students
 - ▶ Plagiarism detection software will be used on all the programming assignments
- ▶ Way of collaboration
 - ▶ You may discuss together or help another student debug code; however, you cannot dictate or give the exact solution



Administrative Stuff

- ▶ Plagiarism punishment
 - ▶ When one student copies from another student, both students are responsible
 - ▶ Zero point on the assignment or exam in question
 - ▶ Repeated violation will result in an F grade for this course as well as further discipline at the school/university level





A brief overview of AI

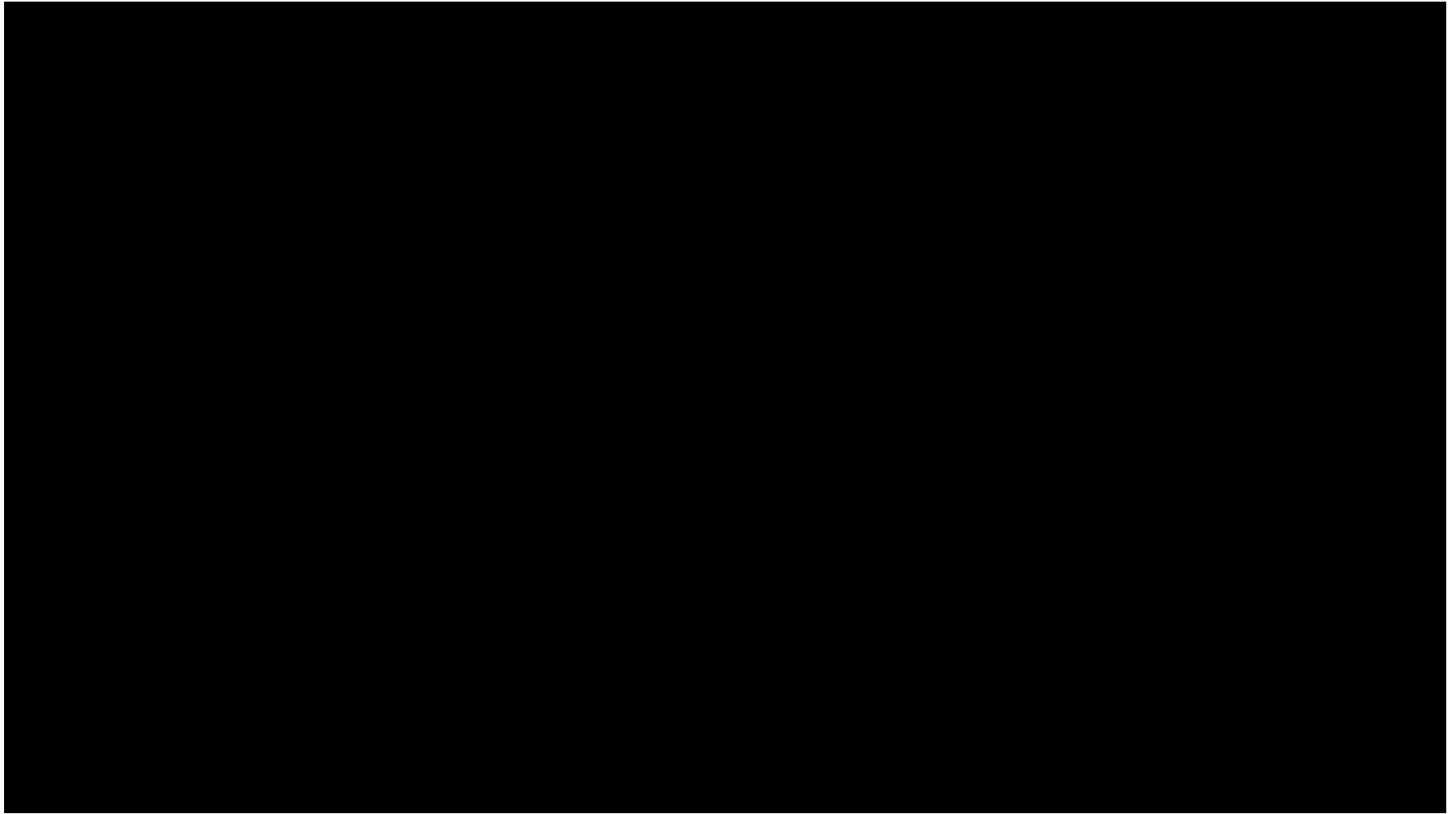


What is artificial intelligence?

▶ AI = Robots?

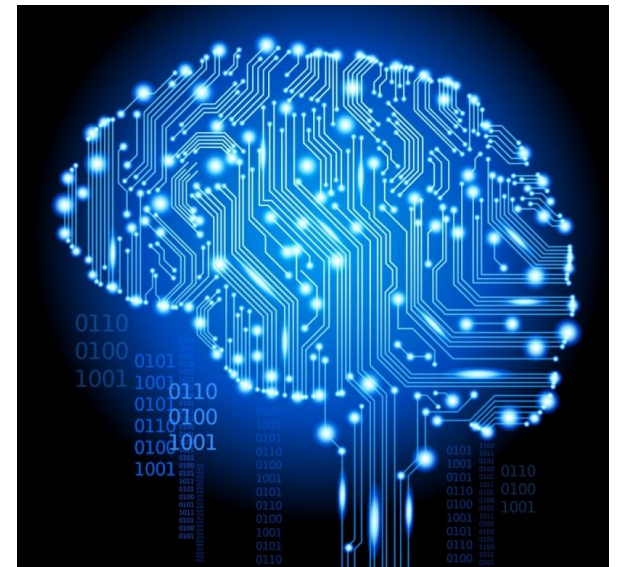


Today's AI 2022->2023->2024->2025



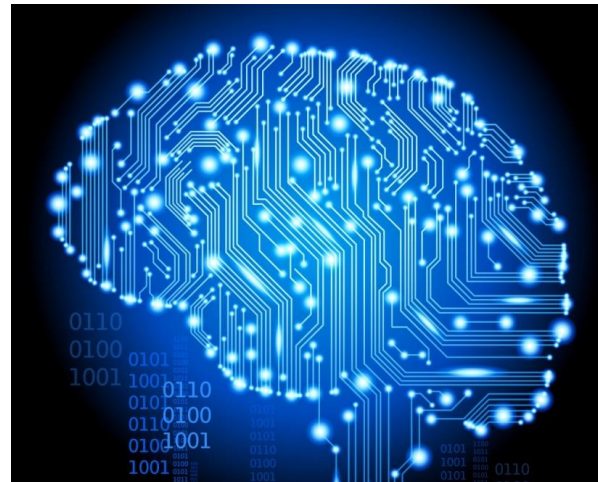
Artificial Intelligence

- ▶ **Artificial intelligence (AI)** is the branch of computer science that develops machines and software with **intelligence**.
- ▶ In this course: intelligence = rationality
- ▶ Rationality: maximize goal achievement given available information (*“do the right thing”*)



Artificial Intelligence

- ▶ AI vs. Human Intelligence
 - ▶ Brains (human minds) are good at rational thinking, but not perfect
 - ▶ “Brains are to intelligence as wings are to flight”



Foundations of AI

- ▶ Philosophy
 - ▶ Logic, reasoning, mind as a physical system, foundations of learning, language and rationality.
- ▶ Mathematics
 - ▶ Formal representation and proof algorithms, computation, (un)decidability, (in)tractability, probability.
- ▶ Psychology
 - ▶ adaptation, phenomena of perception and motor control.
- ▶ Economics
 - ▶ formal theory of rational decisions, game theory.
- ▶ Linguistics
 - ▶ knowledge representation, grammar.
- ▶ Neuroscience
 - ▶ physical substrate for mental activities.
- ▶ Control theory
 - ▶ homeostatic systems, stability, optimal agent design.



A brief history

- ▶ Lots of early speculation & research
 - ▶ Turing: “Computing Machinery and Intelligence” (1950)



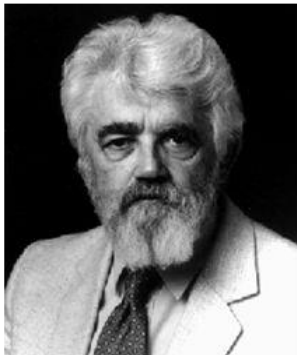
Alan Turing



A brief history

- ▶ Birth (1956)
 - ▶ Dartmouth workshop

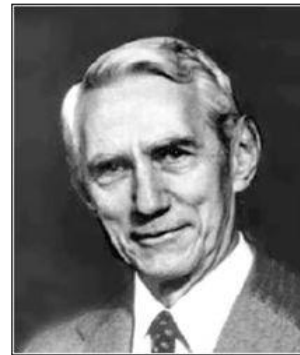
Dartmouth Conference: The Founding Fathers of AI



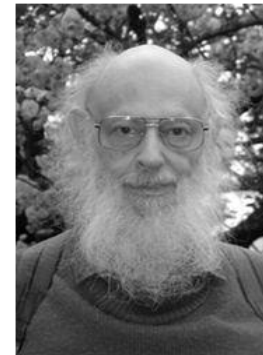
John McCarthy



Marvin Minsky



Claude Shannon



Ray Solomonoff

Alan Newell



Herbert Simon



Arthur Samuel

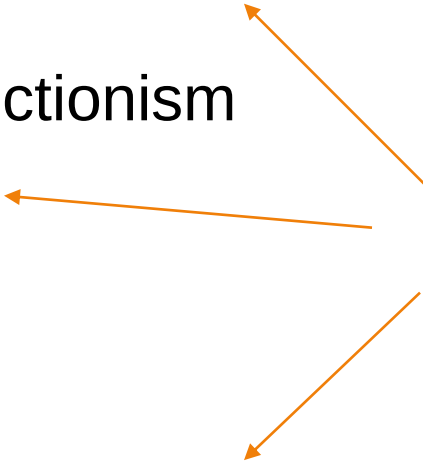


And several
other people...

A brief history

- ▶ Great expectations (1950s-1960s)
 - ▶ A variety of methodology
 - ▶ e.g., symbolism, connectionism
- ▶ AI winter (1970s)
 - ▶ Downfall of perceptron
 - ▶ Lighthill report
- ▶ Boom (1980s)
 - ▶ Revival of neural networks
- ▶ More scientific methods (1990s-2000s)
 - ▶ Statistical approaches

Very rough timeline

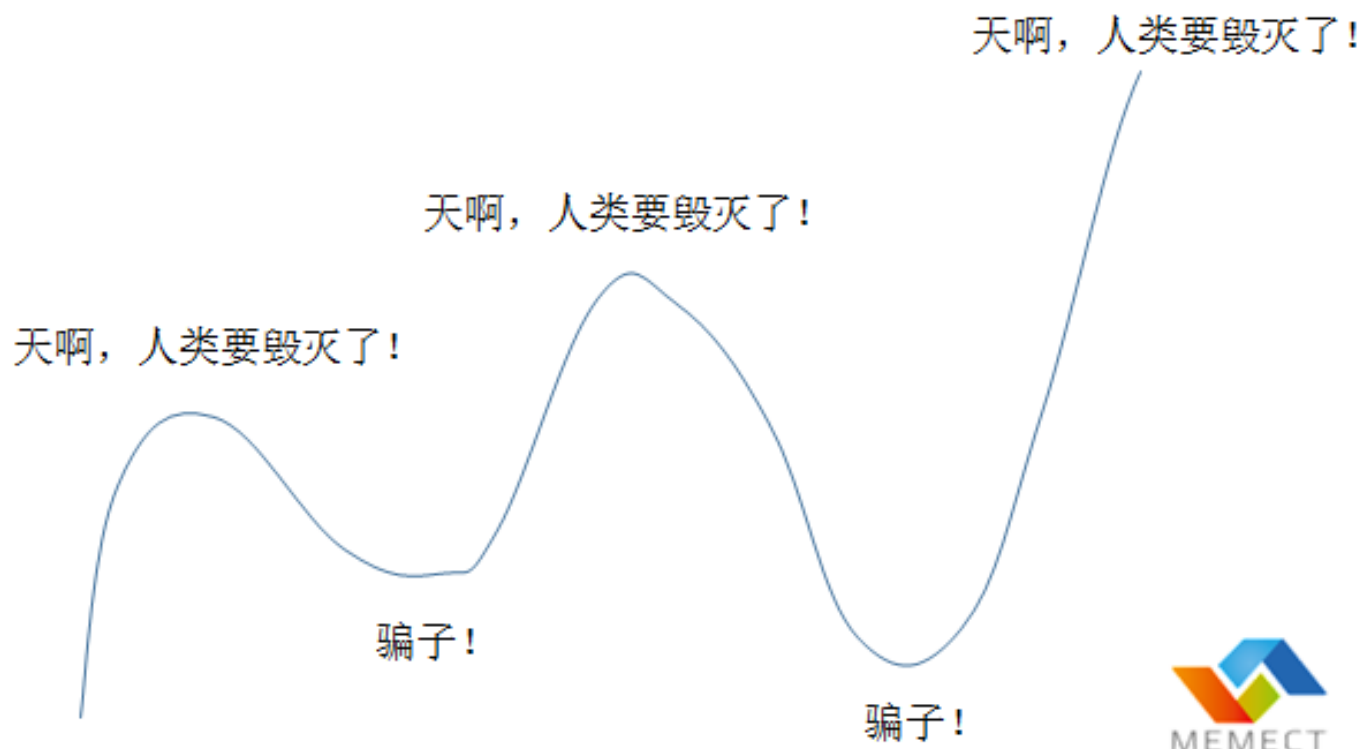


A brief history

- ▶ The past ten years
 - ▶ Rise of big data and deep learning
 - ▶ AI becomes one of the hottest areas in CS
 - ▶ Great interest from industry and public
 - ▶ Many real-world applications



A brief history



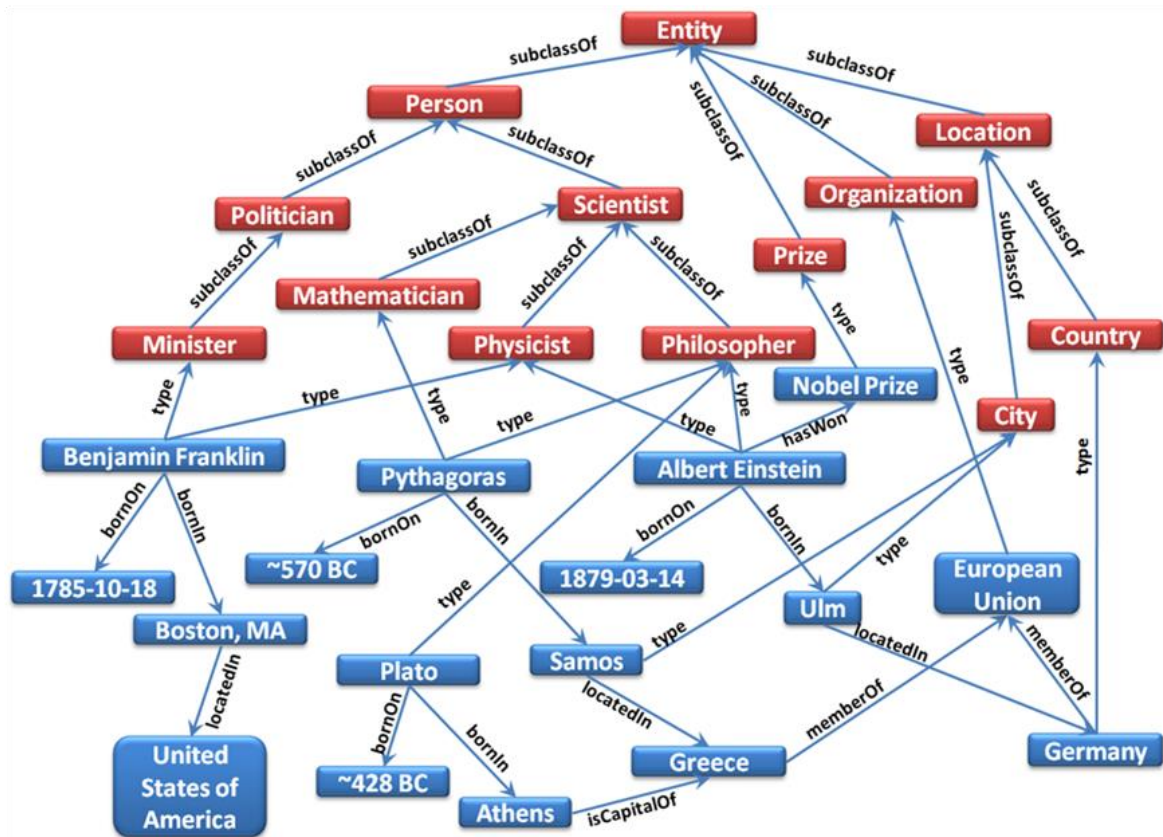
Subfields of AI

- ▶ Many subfields of AI
 - ▶ Knowledge Representation and Reasoning
 - ▶ Knowledge Graphs
 - ▶ Machine Learning
 - ▶ Deep learning
 - ▶ Robotics
 - ▶ Natural Language Processing
 - ▶ Computer Vision
 - ▶ Many more...



Subfields of AI

► Knowledge Representation and Reasoning



Thomas Jefferson

3rd U.S. President

Thomas Jefferson was an American Founding Father, the principal author of the Declaration of Independence, and the third President of the United States. [Wikipedia](#)

Born: April 13, 1743, Shadwell, VA

Died: July 4, 1826, Charlottesville, VA

Presidential term: March 4, 1801 – March 4, 1809

Spouse: [Martha Jefferson](#) (m. 1772–1782)

Party: [Democratic-Republican Party](#)

Awards: AIA Gold Medal

Get updates about Thomas Jefferson

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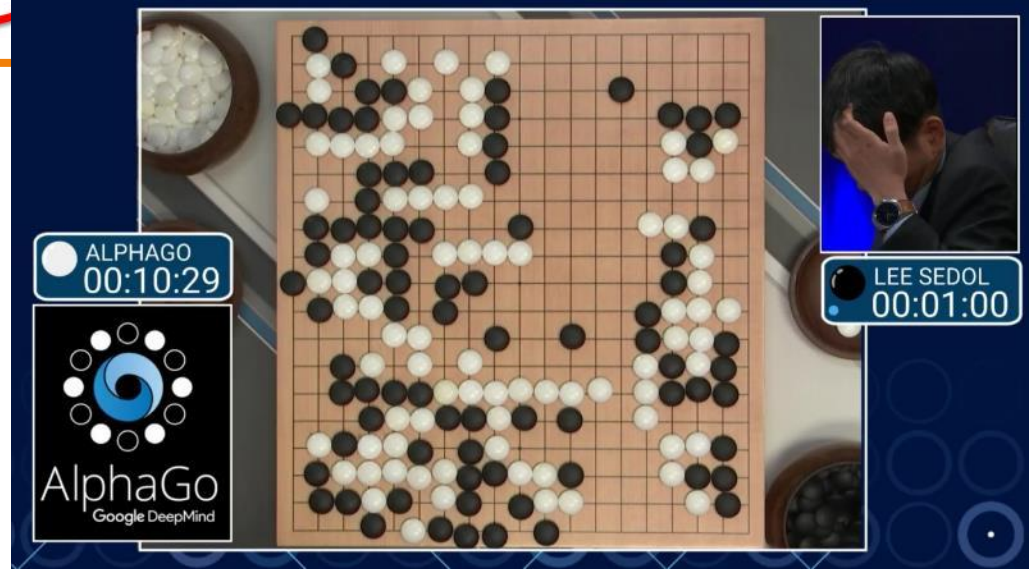
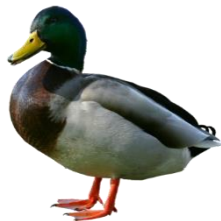
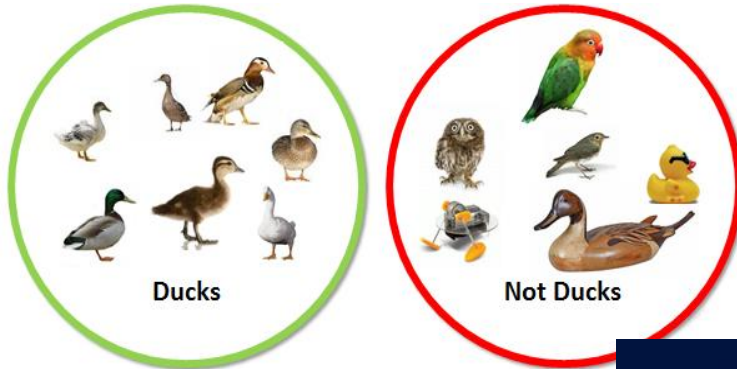


Alexander Hamilton

Feedback

Subfields of AI

► Machine Learning



Subfields of AI

► Robotics



Subfields of AI

► Natural Language Processing

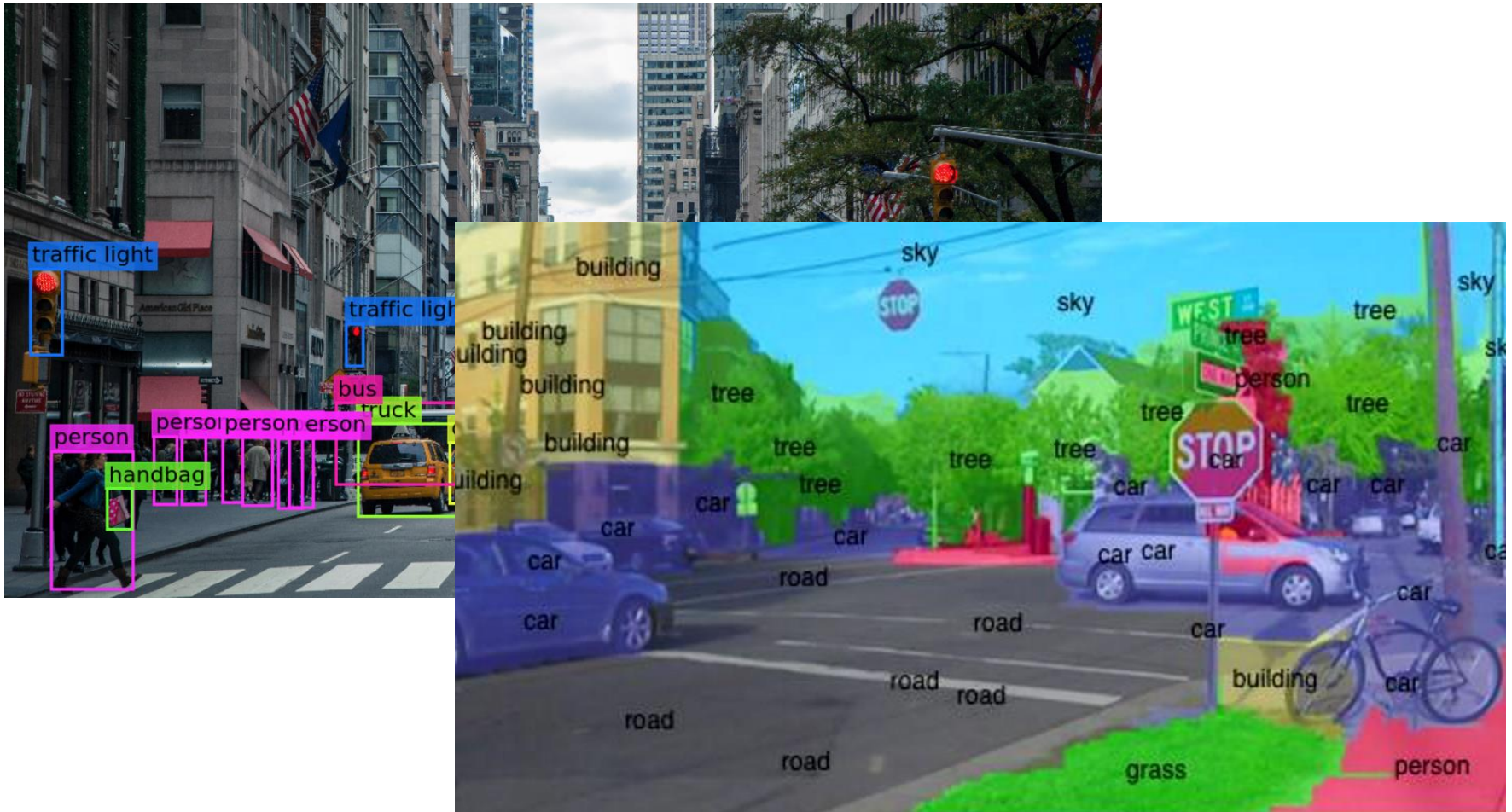


ChatGPT



Subfields of AI

► Computer Vision



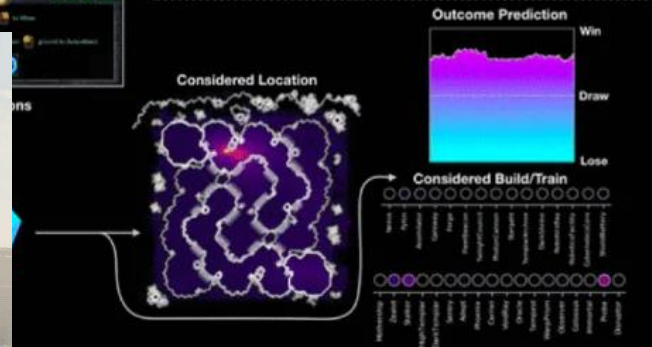
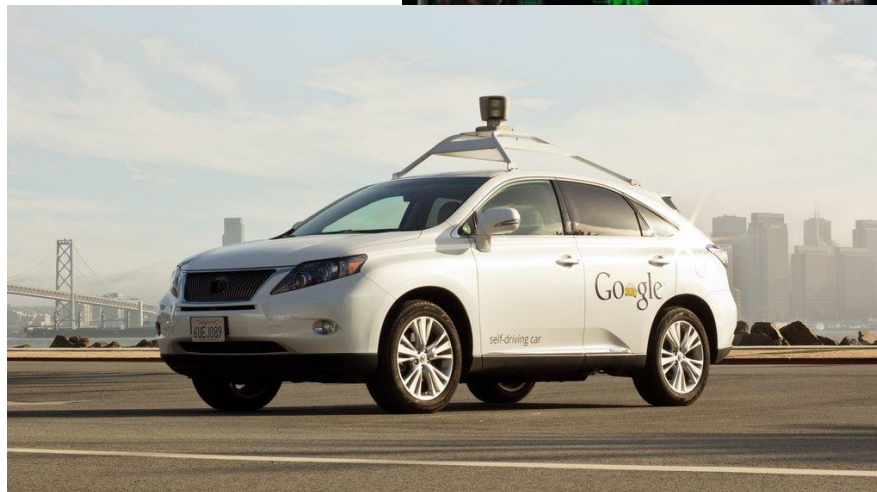
Applications of AI

- ▶ Spam email filter
- ▶ Handwriting recognition
- ▶ Speech recognition
- ▶ Search engine
- ▶ Modern Chinese IME



Applications of AI

- ▶ Financial trading
- ▶ Game AI
- ▶ Customer service chatbot
- ▶ Self-driving



Today's Applications

2024Spring

- ▶ Gemini Demo
- ▶ Sora Demo
- ▶ ...

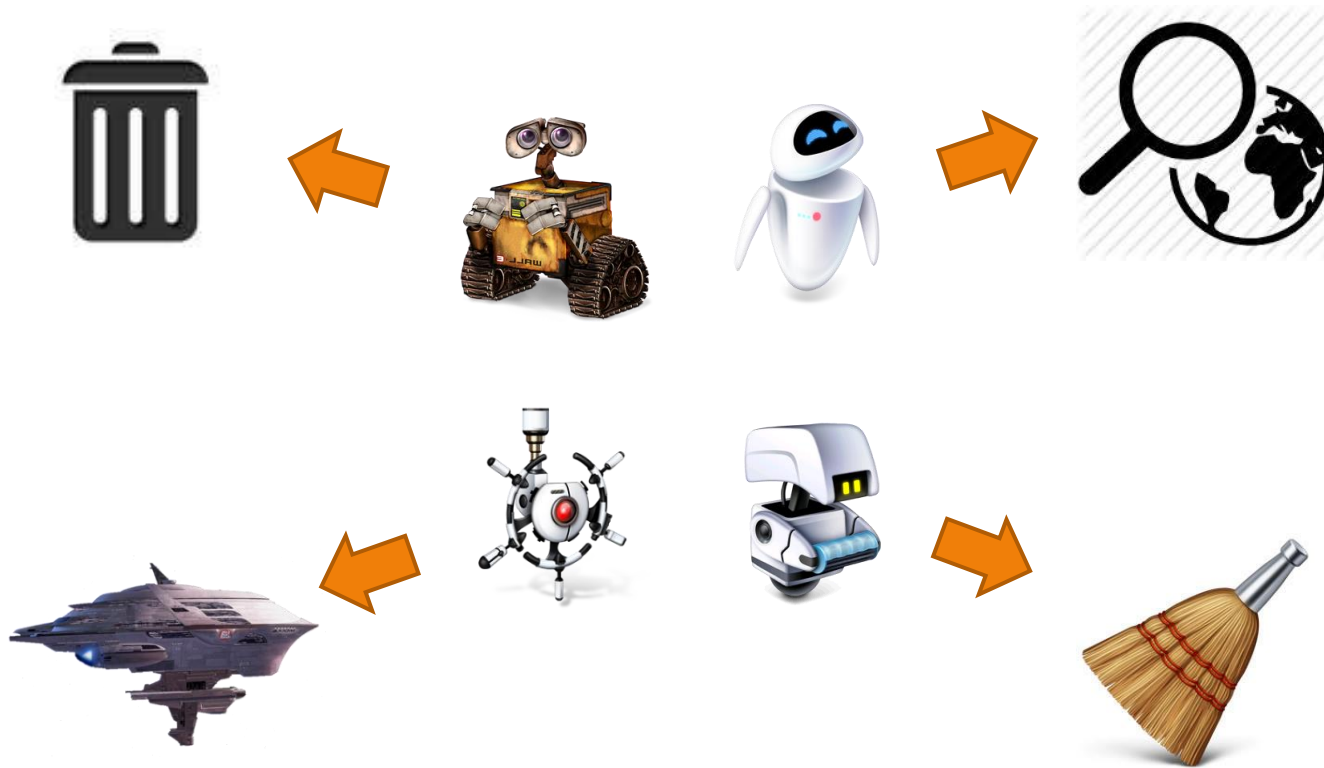
2025Spring

- ▶ Sora/可灵/海螺/混元
- ▶ AI + Robots
- ▶ OpenAI-o1/DeepSeek v3/r1



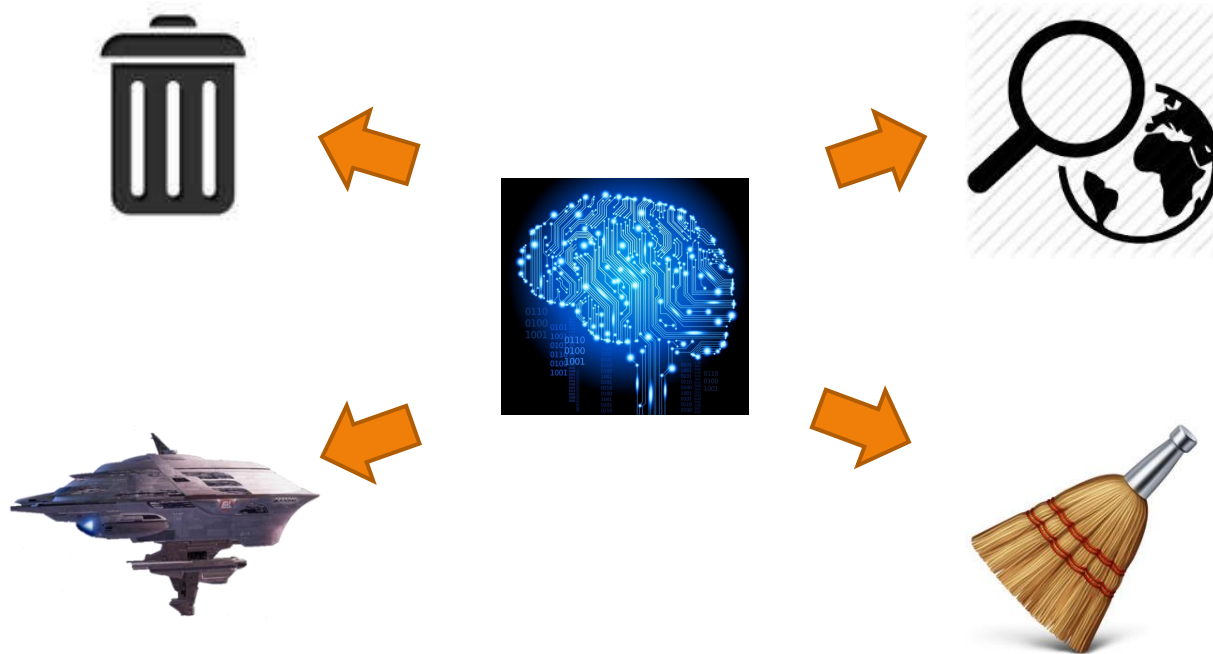
Strong AI vs. Weak AI

- ▶ Weak AI (Applied AI)
 - ▶ AI that accomplishes specific tasks



Strong AI vs. Weak AI

- ▶ Strong AI (General AI)
 - ▶ human-like intelligence – AI that could successfully perform any intellectual task that a human can



Strong AI vs. Weak AI

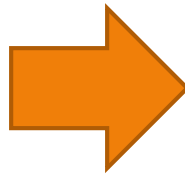


- ▶ Q1: What is the woman in the middle doing?
 - ▶ Action recognition, a CV problem
 - ▶ Q2: What is the woman on the left going to do?
 - ▶ Reasoning about context and intention (beyond current CV)
 - ▶ Q3: This photo was taken in Europe in 2015. What was going on?
 - ▶ Knowing background knowledge by reading news, ...
-



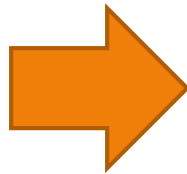
Central problems of (strong) AI

- ▶ Knowledge Representation (KR)
 - ▶ Knowledge: facts, beliefs, concepts, skills, ... that are accumulated over time



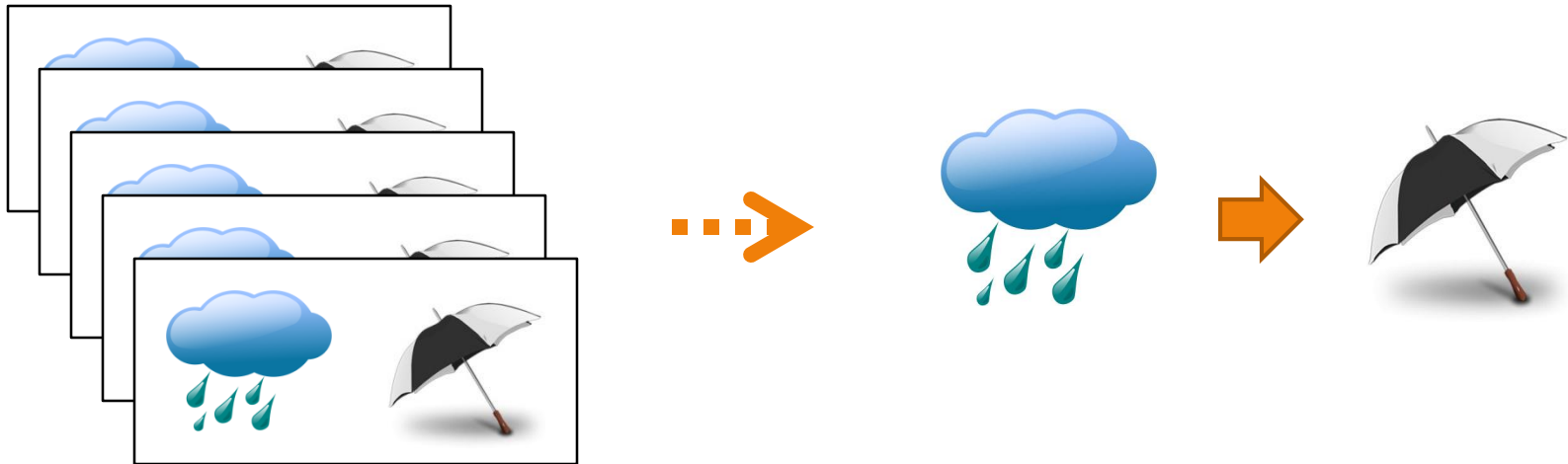
Central problems of (strong) AI

- ▶ Inference
 - ▶ How to utilize knowledge to derive new information based on existing information



Central problems of (strong) AI

- ▶ Learning
 - ▶ How to accumulate knowledge from experience and education



Three types of approaches

- ▶ Symbolism
 - ▶ Physical symbol system hypothesis

“A physical symbol system has the necessary and sufficient means for general intelligent action.”

— Allen Newell and Herbert A. Simon (1976)



Three types of approaches

- ▶ Symbolism
 - ▶ Representing knowledge with symbols and their compositions (expressions)
 - ▶ Inference and learning is done by manipulating symbols (e.g., logic)

$$\begin{aligned} \forall x \forall y, Human(x) \wedge Place(y) \wedge At(x, y) \wedge Rain(y) \\ \rightarrow \exists z, Umbrella(z) \wedge Use(x, z) \end{aligned}$$



Three types of approaches

- ▶ Symbolism

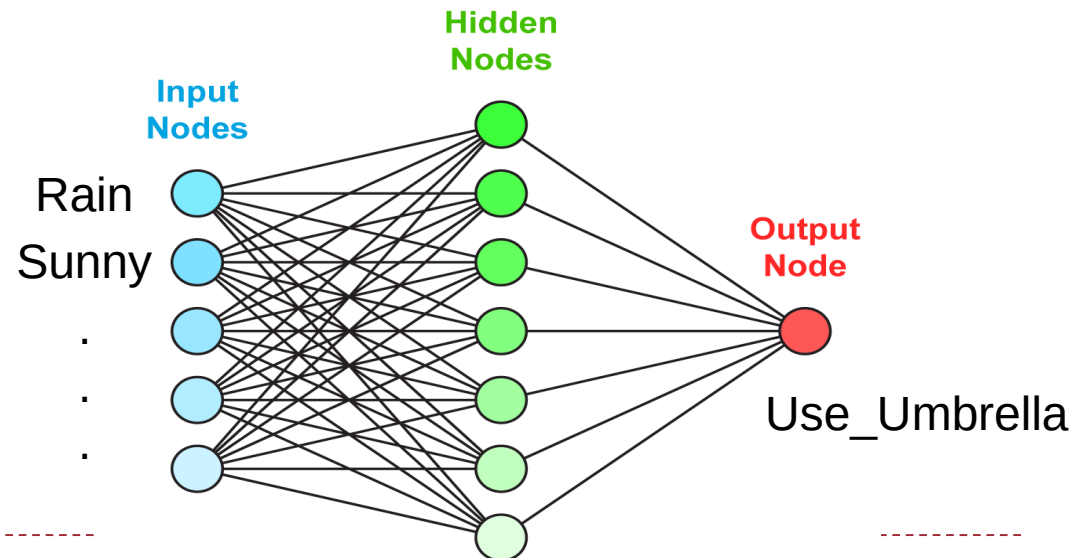
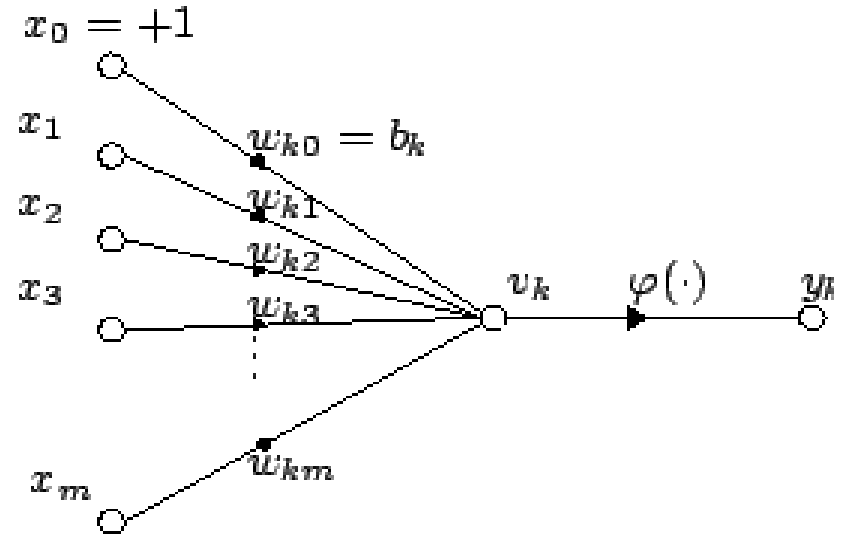
- ▶ History

- ▶ Dominant during 1950s – 1980s
 - ▶ Fell out of favor in 1980s – 1990s
 - ▶ Integration with statistical approaches (2000s)
 - ▶ Integration with neural approaches (2010s)



Three types of approaches

- ▶ Connectionism
 - ▶ Representing knowledge with interconnected networks of simple units
 - ▶ Neural networks



Three types of approaches

- ▶ Connectionism

- ▶ Inference

- ▶ computed based on the functions on the nodes in the network

- ▶ Learning

- ▶ optimization of connection weights

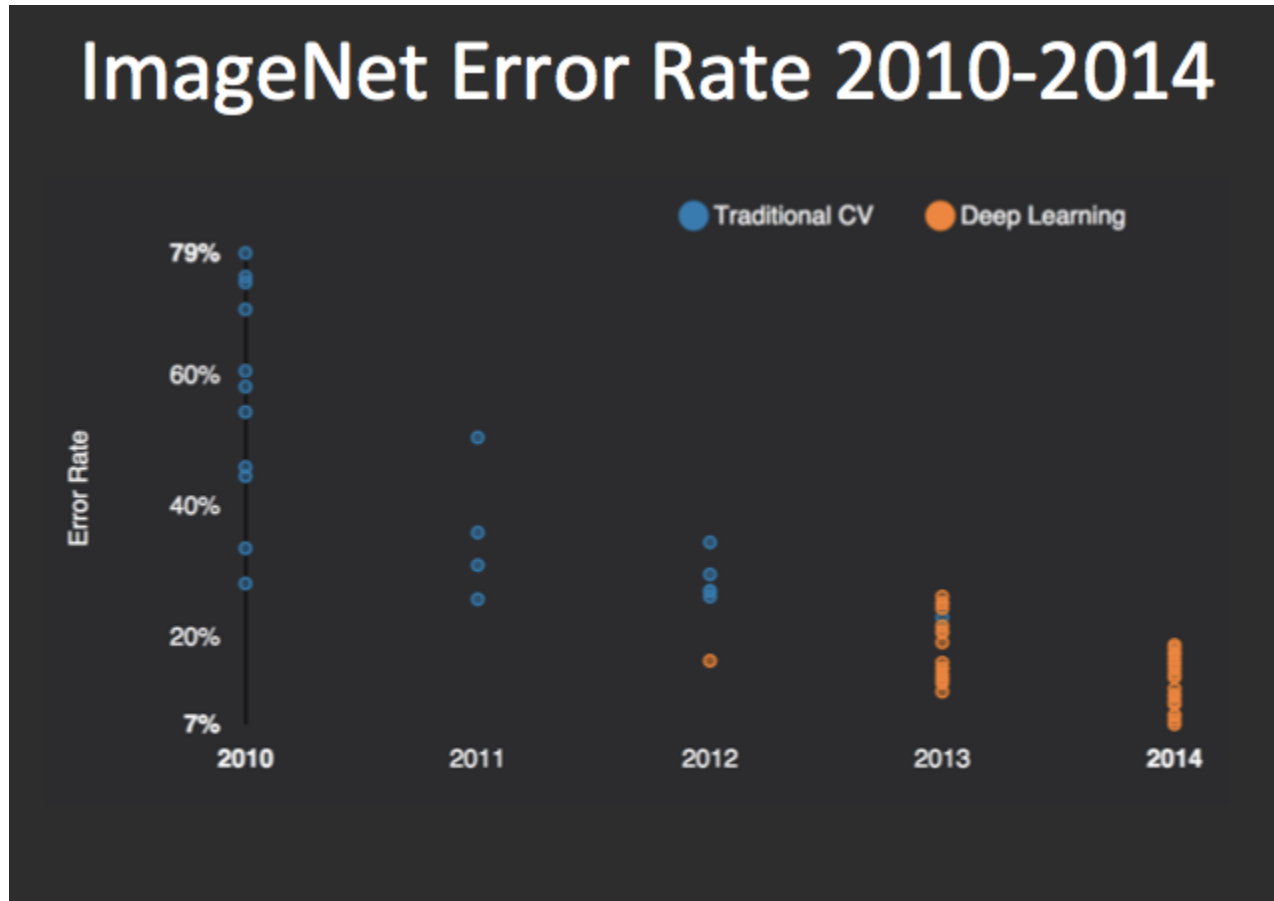


Three types of approaches

- ▶ History of connectionism: rose and fell for several times
 - ▶ 1940s: pioneer work, e.g., McCulloch-Pitts model
 - ▶ 1958: invention of perceptron (Rosenblatt)
 - ▶ 1969: “Perceptron” published (Minsky & Papert)
 - ▶ Publicized key issues of perceptron (e.g., XOR)
 - ▶ 1970s: AI winter
 - ▶ 1980s: revival of connectionism
 - ▶ Hopfield net, BP algorithm
 - ▶ Rumelhart & McClelland (1986): Parallel Distributed Processing
 - ▶ 1990s-2000s: overtaken in popularity by other methods
 - ▶ 2010s: rise of deep learning
 - ▶ Since ~2012: dominates CV
 - ▶ Since ~2015: dominates NLP

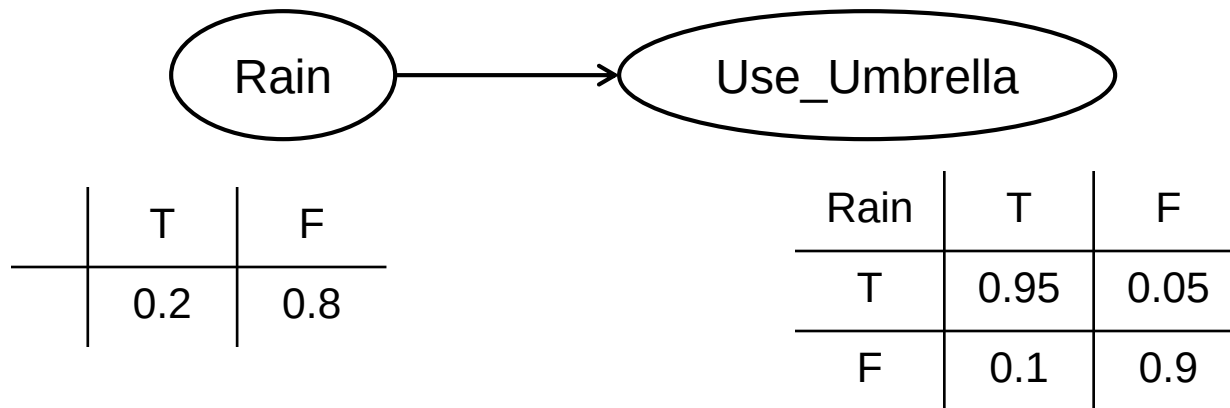


Three types of approaches

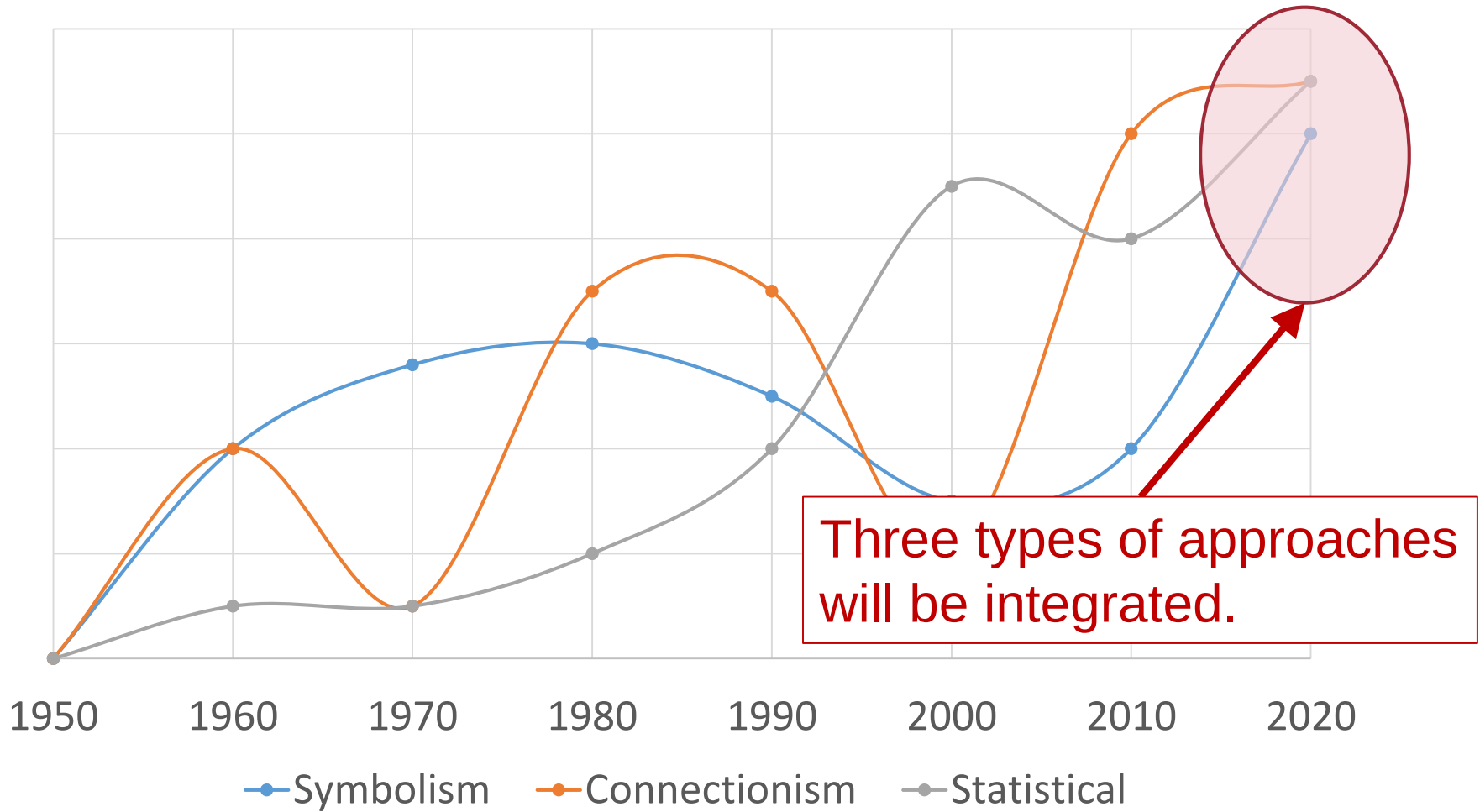


Three types of approaches

- ▶ Statistical approaches
 - ▶ Representing knowledge with probabilistic models
 - ▶ Inference and learning is done by probabilistic inference
 - ▶ History: become popular since 1990s



Trends



Course Overview

- ▶ Search
- ▶ Constraint satisfaction problems
- ▶ Game
- ▶ Propositional logic
- ▶ First-order predicate logic
- ▶ Probabilistic graphical models
- ▶ Probabilistic logics
- ▶ Probabilistic temporal models
- ▶ Markov decision processes and reinforcement learning
- ▶ Introduction to machine learning
- ▶ Introduction to CV, NLP...



Related Courses

▶ Undergraduate

- ▶ SI151 Optimization and machine learning
- ▶ CS150 Database and data mining
- ▶ CS172 Computer vision I

▶ Graduate

- ▶ CS282 Machine Learning
- ▶ CS280 Deep Learning
- ▶ CS272 Computer Vision II
- ▶ CS283 Robotics
- ▶ CS243 Introduction to Algorithmic Game Theory
- ▶ CS284 Simultaneous Localization and Mapping
- ▶ SI251 Convex optimization

